

Assignment: Task Management System (Team Workflow App)

This assignment focuses on building a Task Management System that allows users (or teams) to create, assign, track, and manage tasks efficiently with role-based access and workflow states.

Technologies:

- Frontend: React / Vite.js (TypeScript preferred)
- Backend: Express.js (Node.js)
- Database: MySQL or MongoDB.

Core Functionality:

- **Task Management:**
 - Create a new task with a title, description, priority (Low / Medium / High), due date, and status (Open, In Progress, Done).
 - View all tasks in a list (table or card view).
 - View task details page.
 - Edit, update, and delete tasks.
- **User Management**
 - Register/Login users.
 - Each task should be linked to a user (createdBy / assignedTo).
 - Users can only see their own tasks (basic role separation).
- **Role System**
 - Admin
 - User
 - Admin can see all tasks.
 - Users can only see assigned or created tasks.

Bonus Points:

- User authentication and authorization.
- Designate different task statuses (e.g., "Open," "In Progress," "Testing").
- Search and filtering functionality to filter tasks by title, priority, status, etc.
- Pay extra attention to the user interface and user experience (UI/UX).

Deliverables:

- A functional web application with CRUD operations for tasks.
- Submit your code via a public Git repository.
- Deploy the application to a cloud service (e.g., Vercel, Netlify, AWS) and share a live demo link.
- Include a README file that details setup instructions, dependencies, and usage.

Evaluation Criteria:

- Code quality, readability, and maintainability.
- Completeness of CRUD functionality and user experience.
- Bonus points will be awarded for additional features and robust system design.

Additional Notes:

- You can initially focus on core functionalities and consider adding bonus features later.
- We are interested in your problem-solving approach to data management and user interaction.

Next Steps:

Once you have completed the assignment, submit it as a reply email. We will then schedule a technical interview to discuss your code and the thought process behind the task management application.

This assignment allows you to showcase your skills in building a data-driven application with CRUD operations. We look forward to seeing your work!